RESEARCH BRIEF / MEASURED PILOT

A lunar localization lead and the next decisive test

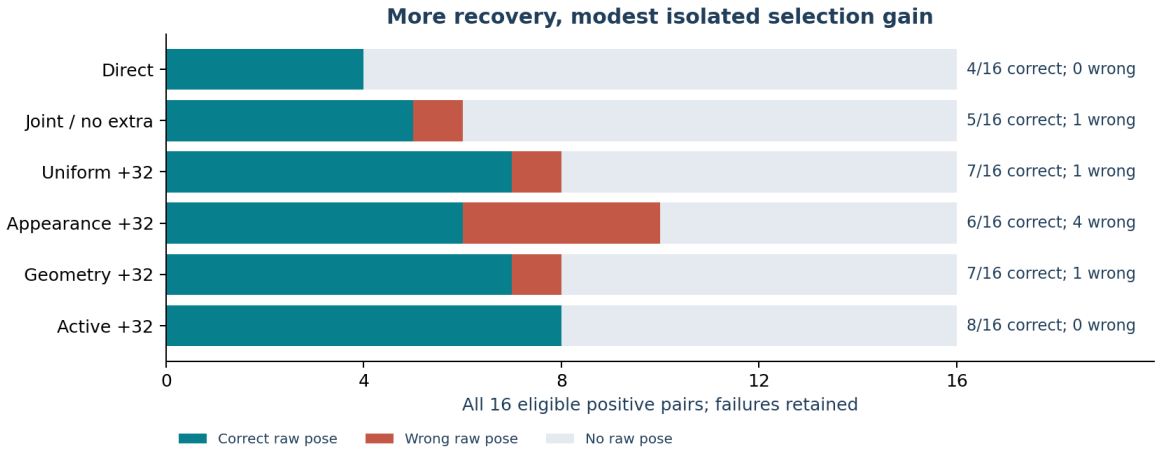
Active patch selection recovers 8/16 raw poses versus 7/16 for the strongest equal-search controls.

What was tested

A classical native-resolution pipeline fits one camera to correlation fields, then sequentially chooses 32 further image measurements. The active score favors patches that distinguish camera hypotheses despite bounded reference-height uncertainty. It adds zero trainable parameters; no learned-model upgrade is demonstrated.

Method	Correct /16	Wrong	Strict /16	Seconds
Direct	4	0	4	1.36
Joint / no extra	5	1	4	7.52
Uniform +32	7	1	4	11.11
Appearance +32	6	4	4	10.55
Geometry +32	7	1	5	11.60
Active +32	8	0	4	11.31

Correct: rotation <=2 degrees and translation <=1% of median reference range. Strict: <=0.5 degrees and <=0.1%. Every eligible positive remains in the denominator; all fresh methods return zero raw poses on 16 nonoverlaps. Seconds are median positive-pair online time on the same CPU.



Interpretation. The isolated policy gain is one pose, or 6.25 points; its descriptive terrain interval is [0,19.58] points. Active is about 8.33x slower than Direct. This is a development lead, not evidence of a superior or efficient model.

What would make the result useful?

The next contribution must survive unknown lighting, stronger controls and independent terrain.

Three limits that matter

The 16 positives reuse four exposed acquisition conditions on three terrain components. Matching reference illumination is supplied. All eight nearly uniform positives still fail; strict precision is better for geometry selection (5/16) than active (4/16). Raw recovery is not calibrated correct acceptance.

The historical 48-positive modern comparison used default-dither RGB8. Fixed detector-coordinate dither can create false image matches in dark renders; the clean control removed three wrong poses without recovering the dark cases. The newer active pilot uses dither-free EXR. These pipelines cannot be pooled into a clean state-of-the-art ranking.

Prior art changes the claim

Ansar and colleagues already demonstrate lunar illumination-aware matching. Active search and learned geometric refinement are established. Lunar MASt3R adaptation and up-to-sevenfold lunar-model distillation also exist. Small size, refinement or lunar adaptation alone is insufficient novelty.

A narrower hypothesis remains worth testing: **selecting actual measurements that distinguish competing camera/light explanations under reference-map uncertainty improves correct camera localization at a fixed search budget.** Establish that gain over uniform and geometry controls before making a learned or novelty claim.

The primary unknown-light study remains unrun

A new reference-only catalogue/camera initializer passes full six-method generated controls. Their pair times are 46.06/51.88 s; the frozen full eight-case forecast is 652.51 s, exceeding its 480 s cap. It stopped before lunar inference. The clean roster contains four positives and four nonoverlaps, with three positive terrain components. It is a diagnostic, not fresh validation.

Original targets remain unchanged: >=70% correctly accepted poses, >=10-point gain with a positive paired terrain interval, <=1% empirical accepted error/nonoverlap acceptance, and a modern accuracy or efficiency advantage. Full A/B/C ablations, seeds, transfer and the untouched final test remain open.

Full 8-page working paper, primary references and source-bound evidence. Measured pilot; clean inputs and cost-gate replay. No email or paper submission has been made.